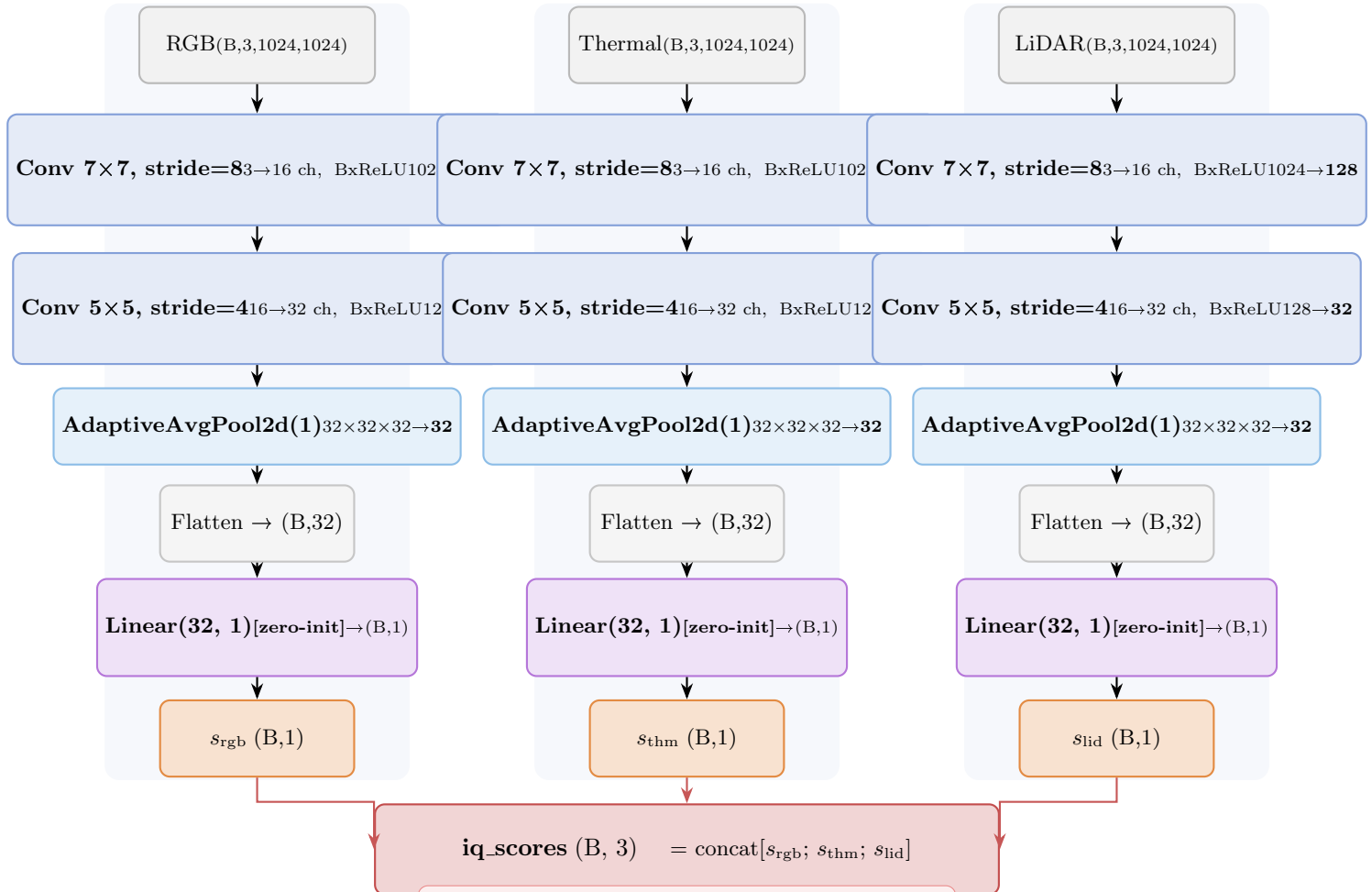


InputQualityEstimator (IQE) — Phase 0 [Fully Learnable, End-to-End Backprop]



Backpropagation paths through IQE:

(1) **Direct:** $\mathcal{L}_{\text{iq_kl}} = D_{\text{KL}}(\hat{w} \parallel \text{softmax}(\text{iq_scores})) \rightarrow$
 gradients flow into $s_i \rightarrow \text{Linear} \rightarrow \text{Conv}$ layers

Why CNN instead of handcrafted features?

Previous (bad): brightness, Laplacian, SNR $\rightarrow$ heuristic
 $\Rightarrow$ @no-grad on features, only MLP trained
 $\Rightarrow$ quality metrics fixed by human assumption

Current (good): CNN learns what signals matter for quality in each sensor type:

- RGB CNN: learns night brightness pattern
- Thermal CNN: learns thermal contrast
- LiDAR CNN: learns spatial density

Full end-to-end training via KL + seg losses.

direct: seg loss $\rightarrow$ fused_output $\rightarrow$ AMF weights
 $= \text{feat_logits} + \text{iq_scale} \cdot \text{iq_scores} \rightarrow \text{IQE}$

oracle $\hat{w}$ is .detach()'d (correct — it is the target earned path)

Architecture summary:

input: (B, 3, 1024, 1024)
 Conv7x7, s=8: (B,16,128,128)
 Conv5x5, s=4: (B,32,32,32)
 GAP: (B,32,1,1)
 Flatten: (B,32)
 Linear(32,1): (B,1)

$\approx 15\text{K}$ params/modality
 $\times 3$ modalities = **45K total**

zero-init Linear: ensures uniform weights at init